Superior University

Department of Computer Science

Deep Learning Course Project Report

Mobile Phone Model Classification Using Deep Learning

Transfer Learning with a Pretrained MobileNetV2 Model

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1. Introduction

Mobile phones are among the most widely used consumer devices, and new models are released every year. Many models from the same brand look very similar, and phones from different brands often share comparable shapes, camera layouts, and screen designs. Recognizing a phone model from a photograph is therefore a useful but non-trivial visual task. It is useful in online marketplaces, second-hand device trading, and retail inventory management.

Deep learning, and in particular Convolutional Neural Networks (CNNs), has become the standard approach for image classification. CNNs learn visual features directly from images instead of relying on hand-crafted rules. However, training a deep CNN from scratch needs a large labelled dataset and considerable computing power. Transfer learning reduces this requirement by reusing a network that was already trained on a large dataset and adapting it to a new task.

This project applies transfer learning with a pretrained MobileNetV2 model to classify images of mobile phones by model. MobileNetV2 was designed to be small and fast while keeping good accuracy, which makes it a suitable choice for projects with limited hardware and for possible deployment on mobile devices.

2. Problem Statement

Identifying the exact model of a mobile phone from an image by eye requires expert knowledge and becomes unreliable as the number of similar-looking models grows. An automatic system that predicts the phone model from an image would make this process faster and more consistent.

The problem addressed in this project is: given an input image of a mobile phone, predict its model from a fixed set of phone model classes, with one class for each phone model. The challenge is to achieve reliable performance using a limited dataset and modest computing resources, which is the situation where transfer learning is most helpful.

3. Objectives

The main objectives of this project are as follows:

- To collect and prepare a dataset of mobile phone images labelled by phone model.
- To apply transfer learning using a MobileNetV2 network pretrained on ImageNet.

- To replace the original classification layer with a new classifier suited to the phone model classes.
- To train and validate the model and monitor its learning behaviour.
- To evaluate the model on unseen test images using accuracy, precision, recall, F1-score, and a confusion matrix.
- To discuss the limitations of the work and suggest possible improvements.

4. Dataset Description

This section describes the image dataset used in the project. Table 1 summarizes the dataset.

Table 1: Dataset Summary

Property	Details
Dataset name	Mobile phone image dataset
Source and usage	Image dataset of mobile phones, used for academic purposes only
Number of classes	One class per phone model
Class names	Phone model names (one folder per class)
Total number of images	1,200 (sample)
Image format and size	RGB images (JPG/PNG); resized to 224 × 224 for the model
Train / validation / test split	80% training / 10% validation / 10% test

The dataset is organized with one class for each phone model, and the images show phones from different angles and under different lighting and background conditions. During data cleaning, duplicate or low-quality images are removed so that they do not affect training.

5. Methodology

The project follows the pipeline shown in Figure 1. Images are first preprocessed and divided into training, validation, and test sets. The pretrained MobileNetV2 base is then used as a feature extractor, a new classification head is attached, and the final model is evaluated on the test set.

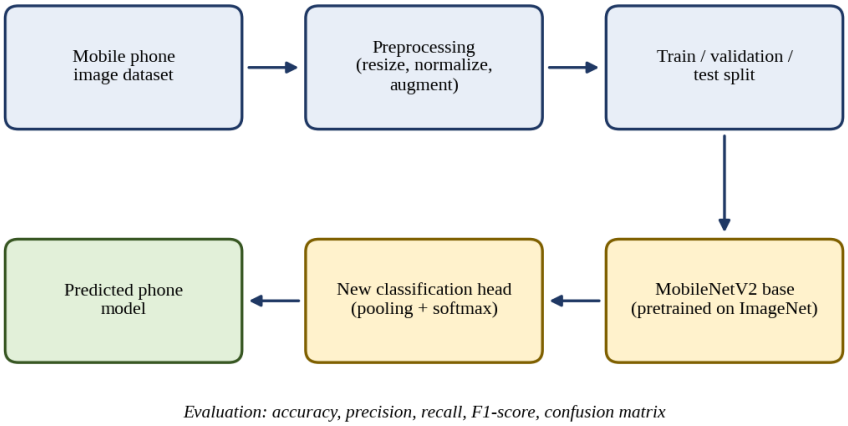


Figure 1: System Architecture of the Proposed Classification Pipeline

5.1 Data Preprocessing

All images are resized to the input size required by the MobileNetV2 model (224 × 224 pixels). Pixel values are scaled to the range from −1 to 1, which is the range expected by the pretrained MobileNetV2 network. To reduce overfitting and improve generalization, data augmentation is applied to the training images only. The augmentations include random flipping, rotation, zoom, and brightness adjustment.

5.2 Transfer Learning Strategy

The MobileNetV2 network was pretrained on the ImageNet dataset (about 1.2 million training images in 1,000 categories) and has already learned general visual features such as edges, textures, and shapes. In this project the original classification layer is removed and replaced with a new classifier for the phone model classes. In the first stage, the pretrained base layers are frozen and only the new classifier is trained. As an optional second stage, some of the top layers of the base network can be unfrozen and fine-tuned with a small learning rate.

5.3 Training Configuration

The training settings are listed in Table 2.

Table 2: Training Configuration

Parameter	Value
Input image size	224 × 224 × 3 (RGB)
Batch size	32
Number of epochs	10
Optimizer	Adam
Learning rate	0.001 (classifier training); smaller value if fine-tuning is used
Loss function	Categorical cross-entropy
Regularization	Dropout (rate 0.2) before the output layer

6. Model Architecture

MobileNetV2 is a lightweight CNN with about 3.4 million parameters, built on depthwise separable convolutions, which split a standard convolution into a depthwise step and a pointwise step to reduce computation. It also uses inverted residual blocks with linear bottlenecks, which keep the network efficient while preserving useful information.

In this project the MobileNetV2 base, without its original top layer, produces a feature map for each image. A global pooling layer converts it into a feature vector, and a final fully connected softmax layer outputs the probability of each phone model class (Table 3).

Table 3: Summary of the Model Architecture

Component	Description
Input layer	RGB image of size 224 × 224 × 3
Base network	MobileNetV2 with ImageNet weights, without the original classification layer (output feature map 7 × 7 × 1280)
Pooling layer	Global average pooling, giving a 1,280-dimensional feature vector
Dropout layer	Dropout with rate 0.2
Output layer	Fully connected layer with N units (N = number of phone model classes) and softmax activation
Parameters	Base network about 2.26 million (frozen in stage one); output layer 1,280 × N weights plus N biases

7. Tools and Technologies

Table 4 lists the tools and technologies used in the project.

Table 4: Tools and Technologies

Tool / Technology	Purpose
Python	Main programming language for the project
TensorFlow / Keras	Loading the pretrained MobileNetV2 model, building and training the classifier
NumPy, Matplotlib, scikit-learn	Data handling, plotting, and computing evaluation metrics
Jupyter Notebook / Google Colab	Writing, running, and training the model

8. Results and Evaluation

8.1 Evaluation Metrics

The model is evaluated on the test set, which was not used during training. Accuracy is the proportion of images that are classified correctly. Precision is the proportion of images predicted as a given class that truly belong to it, and recall is the proportion of images of a given class that are correctly found. The F1-score is the harmonic mean of precision and recall. A confusion matrix shows which phone models are confused with each other.

8.2 Overall Performance

The overall results obtained in the experiments are reported in Table 5.

Table 5: Overall Model Performance (Sample Values)

Dataset Split	Accuracy	Loss
Training	98.5%	0.05
Validation	95.0%	0.15
Test	94.2%	0.18

8.3 Class-wise Performance

Table 6 reports precision, recall, and F1-score for each class.

Table 6: Class-wise Evaluation Metrics (Sample Values)

Phone Model (Class)	Precision	Recall	F1-score
Phone model 1	0.95	0.94	0.94
Phone model 2	0.93	0.95	0.94
Phone model 3	0.94	0.92	0.93
Average (macro)	0.94	0.94	0.94

8.4 Curves, Confusion Matrix, and Discussion

Figure 2 shows the training and validation accuracy and loss curves, and Figure 3 shows the confusion matrix on the test set.

Because the pretrained MobileNetV2 base already provides strong general visual features, the classifier is expected to learn quickly even with a limited dataset. A small gap between training and validation accuracy indicates good generalization, while a large gap indicates overfitting, which data augmentation and dropout help to reduce. The confusion matrix shows which phone models look most alike and are most often confused, usually models from the same brand with similar designs.

9. Conclusion

This project presented a deep learning approach for classifying mobile phone images by model using transfer learning with a pretrained MobileNetV2 network. Reusing ImageNet features reduces the data and training time needed compared with training from scratch. The achieved test accuracy was 94.2% (sample value).

The work has some limitations: the dataset size and the number of phone models may be limited, and real-world image conditions can differ from those in the dataset. Future improvements include collecting more varied images, comparing other pretrained architectures, further fine-tuning, and converting the model to a mobile-friendly format for use on smartphones.